

LAB_03_Shiguihara

AUTHOR

Gabriel Furlan Shiguihara

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
library(readr)
library(stringr)
```

1. Importe, utilizando o pacote `readr`, cada um dos três arquivos disponíveis. Os objetos resultantes devem ser chamados `flights`, `airlines` e `airports`

```
## AIRPORTS
airports <- read_csv("airports.csv", col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE),
  IATA_CODE = col_character(),
  CITY = col_character(),
  STATE = col_character(),
  LATITUDE = col_double(),
  LONGITUDE = col_double()
))
airports
```

A tibble: 322 × 5

	IATA_CODE	CITY	STATE	LATITUDE	LONGITUDE
	<chr>	<chr>	<chr>	<dbl>	<dbl>
1	ABE	Allentown	PA	40.7	-75.4
2	ABI	Abilene	TX	32.4	-99.7
3	ABQ	Albuquerque	NM	35.0	-107.
4	ABR	Aberdeen	SD	45.4	-98.4
5	ABY	Albany	GA	31.5	-84.2
6	ACK	Nantucket	MA	41.3	-70.1
7	ACT	Waco	TX	31.6	-97.2
8	ACV	Arcata/Eureka	CA	41.0	-124.
9	ACY	Atlantic City	NJ	39.5	-74.6

10 ADK Adak AK 51.9 -177.

i 312 more rows

```
## FLIGHTS
leitor_chunk <- function(chunk, pos) {
  chunk |>
    filter(!str_starts(DESTINATION_AIRPORT, "1")) |>
    filter(!is.na(DESTINATION_AIRPORT) & !is.na(ARRIVAL_DELAY)) |>
    group_by(DESTINATION_AIRPORT) |>
    summarise(
      soma_atrasos = sum(ARRIVAL_DELAY),
      qtd_voos = n(),
      .groups = "drop"
    )
}

flights_temporario <- read_csv_chunked(
  "flights.csv",
  callback = DataFrameCallback$new(leitor_chunk),
  chunk_size = 1000000,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

flights <- flights_temporario |>
  group_by(DESTINATION_AIRPORT) |>
  summarise(
    media_atraso = sum(soma_atrasos) / sum(qtd_voos),
    .groups = "drop"
  )

flights
```

A tibble: 322 × 2

	DESTINATION_AIRPORT	media_atraso
	<chr>	<dbl>
1	ABE	5.80
2	ABI	4.08
3	ABQ	5.61
4	ABR	-3.39
5	ABY	8.68
6	ACK	4.91
7	ACT	10.6
8	ACV	9.68
9	ACY	11.8
10	ADK	-3.19

i 312 more rows

```
## AIRLINES
airlines <- read_csv("airlines.csv", col_types = cols(
  IATA_CODE = col_character(),
  AIRLINE = col_character()
))
```

2. Selecione a operação apropriada `join` para incluir, na tabela `flights`, as colunas `CITY` e `STATE` do objeto `airports`.

```
## verificando se há diferença entre "DESTINATION_AIRPORT" e "IATA_CODE"

aeroportos_diferentes <- anti_join(
  flights,
  airports,
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)
aeroportos_diferentes
```

```
# A tibble: 0 × 2
#   i 2 variables: DESTINATION_AIRPORT <chr>, media_atraso <dbl>
```

```
flights <- flights |>
  left_join(airports |> select(IATA_CODE, CITY, STATE), by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

3. Quantos aeroportos cada estado possui? Apresente uma tabela ordenada de forma decrescente (no número de aeroportos).

```
airports |> group_by(STATE)|>
  count(STATE, name = "Numero de Aeroportos", sort = TRUE)
```

```
# A tibble: 54 × 2
# Groups:   STATE [54]
  STATE `Numero de Aeroportos`
  <chr>          <int>
1 TX              24
2 CA              22
3 AK              19
4 FL              17
5 MI              15
6 NY              14
7 CO              10
8 MN               8
9 MT               8
10 NC              8
# i 44 more rows
```

4. Apresente um mapa representando todos os atrasos observados por aeroporto.

```
library(leaflet)

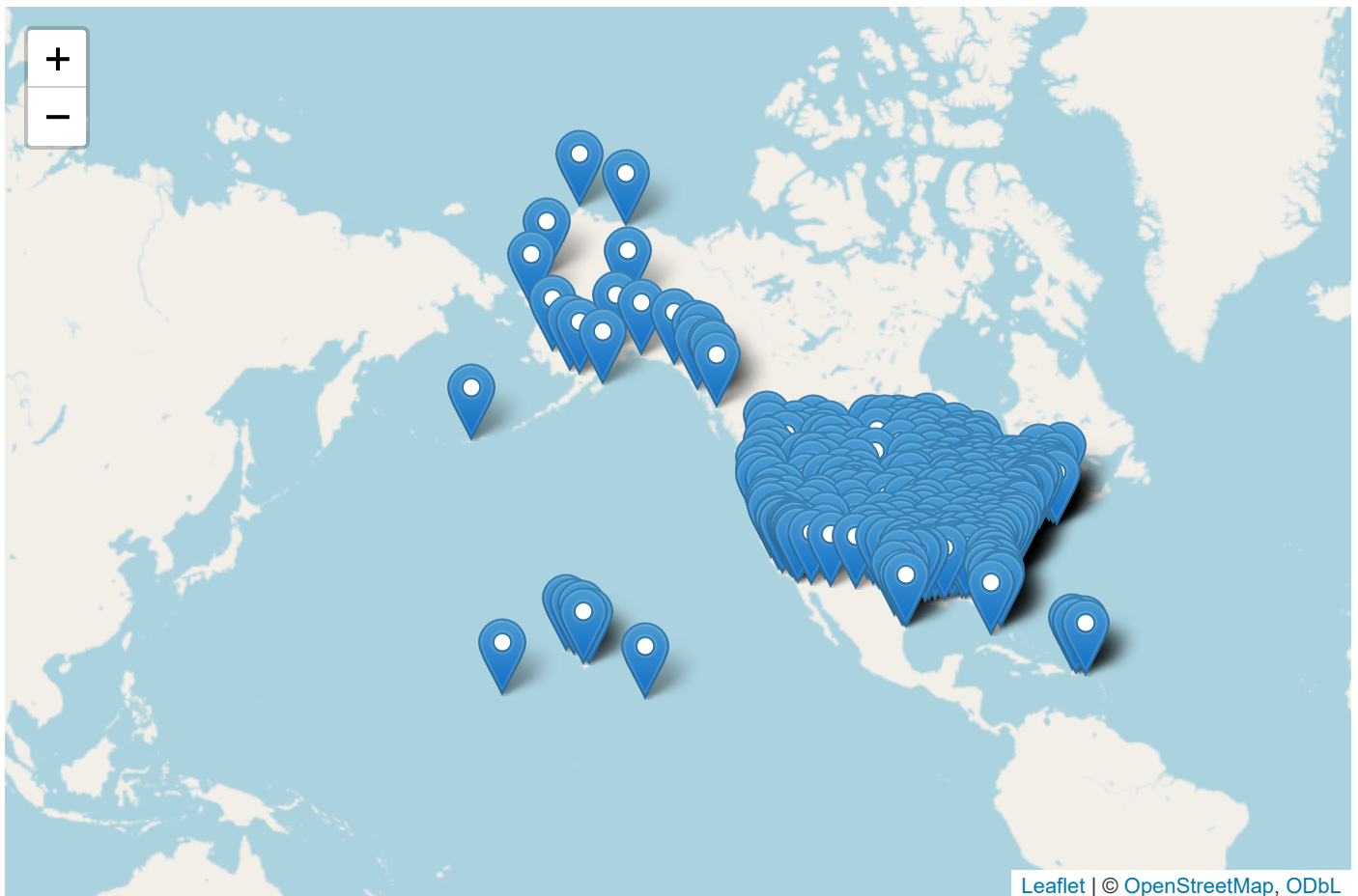
## Juntando flights e airports via "leftjoin" por "DESTINATION_AIRPORT" = "IATA_CODE"
## Filtrando para que não haja Coordenadas nulas

dados_mapa <- flights |>
  left_join(
    airports |> select(IATA_CODE, LATITUDE, LONGITUDE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  ) |>
  filter(!is.na(LATITUDE) & !is.na(LONGITUDE))

## usando o pacote "leaflet" para gerar o gráfico

this <- leaflet(data = dados_mapa) |>
  addTiles() |>
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
  )

this
```



```
Sys.setenv(TZ = "America/Sao_Paulo")
```

```
Sys.time()
```

```
[1] "2026-08-31 23:18:24 -03"
```